

# The Income You Can't See:

## Machine Learning, Measurement Error, and the Limits of Survey Correction in the Informal Sector

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### Abstract

Standard household surveys measure self-employment income as conventional profit — revenue minus purchased inputs — ignoring the opportunity cost of the owner's labour. This paper exploits a unique linked dataset of 500 Bolivian informal firms observed simultaneously by a seven-day financial diary and the national household survey to quantify the resulting measurement error and test whether machine learning can recover the true income distribution without diary data. Three results emerge. First, a negative result: unsupervised methods — isolation forests and autoencoders — cannot correct the error because the identifying assumption required for transfer learning fails. The income-covariate relationship differs structurally between salaried and self-employed workers (correlation =  $-0.268$ ), so models trained on salaried workers do not transfer. The autoencoder reduces RMSE by 66% but achieves near-zero rank correlation ( $\rho = 0.023$ ), compressing all incomes toward the mean without recovering rank order — a consequence I formalize as an impossibility result under assumption failure. Second, a modest positive result: the autoencoder improves classification of “trapped” firms (Type C) — precision rises from 37.8% to 43.6%, recall from 20.8% to 34.2%. Third, a constructive result: a supervised gradient boosted regressor trained on 500 diary observations achieves  $\rho = 0.576$  for correction and 73% correct sign, substantially outperforming the linear accounting identity ( $\rho = -0.060$ ). Among the 500 diary firms, the poverty headcount rises from 0.296 (survey) to 0.874 (diary truth). The cost of the enabling diary is approximately \$40,000 — a modest investment for correcting a 14,000-household survey. **Keywords:** measurement error, informal

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sector, machine learning, self-employment income, poverty measurement, Bolivia **JEL**

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# 1 Introduction

How much do informal workers really earn? The question matters for poverty measurement, tax policy, social protection design, and the ongoing debate over whether informality reflects a voluntary choice or an involuntary trap. Yet the answer depends almost entirely on a single survey item — self-reported income or profit — whose accuracy for the self-employed has been questioned for decades. De Mel et al. (2009) show that alternative profit measures for microenterprises disagree by 30–100% depending on the question asked; Hurst et al. (2014) demonstrate systematic underreporting using expenditure-based proxies in the United States.

The problem documented here is distinct from underreporting. Household surveys typically ask self-employed respondents to report “net income” or “profit,” which respondents interpret as conventional profit: revenue minus purchased inputs. This measure systematically omits the opportunity cost of the owner’s labour — the hours the entrepreneur could have sold on the wage market. The omission is not a survey design flaw; it reflects a conceptual mismatch between the accounting profit that respondents can observe and the economic profit that welfare analysis requires. The resulting measurement error is large, positive, mean-reverting, and heteroskedastic — a structure that violates every assumption of classical measurement error theory (Bound et al., 2001).

This paper asks whether machine learning can recover the true income distribution from standard survey data, without requiring the expensive financial diaries that would provide ground truth. The answer, in short, is: not without supervision.

I exploit a unique linked dataset from Bolivia. Five hundred urban self-employed firms were observed simultaneously by two instruments: the national household survey (Encuesta de Hogares, EH) and a seven-day double-entry financial diary administered by trained enumerators. The diary records revenue, all input costs, owner hours, family labour hours, and asset depreciation — everything needed to compute economic profit,  $\pi^{\text{adj}}$ . The EH records the same firms’ self-reported conventional profit,  $y^{\text{EH}}$ . The linked data permit exact, firm-level measurement of the error  $\varepsilon_i = y_i^{\text{EH}} - \pi_i^{\text{adj}}$ .

Three results emerge.

*First, a negative result.* Unsupervised machine learning — isolation forests for anomaly detection, autoencoders for reconstruction-based correction — cannot recover the measurement error structure. The identifying assumption required for unsupervised transfer fails: the conditional distribution of income given covariates differs structurally between

salaried workers (the training population) and the self-employed (the target population). I formalize this as Proposition 1: when the conditional invariance assumption fails, any reconstruction-based corrector converges to the conditional mean of the training population, compressing the target distribution and destroying rank information. Empirically, the autoencoder reduces RMSE from 5,287 to 1,777 Bs/week (a 66% reduction) but achieves a Spearman rank correlation of only +0.023 against diary truth. The isolation forest performs at or below chance ( $AUC = 0.46\text{--}0.50$ ) across all hyperparameter configurations.

*Second, a modest positive result.* Despite its inability to recover levels, the autoencoder does improve classification of firm types defined in the companion paper (Hernani-Limarino, 2025a). For Type C firms — those trapped in self-employment because their opportunity cost exceeds their economic profit but whose conventional profit appears positive — precision rises from 37.8% to 43.6% and recall from 20.8% to 34.2%. This improvement is not transformative, but it is the best available without diary data.

*Third, a constructive result.* A supervised gradient boosted regressor trained on the 500 diary observations achieves  $AUC = 0.998$  for error detection and Spearman  $\rho = 0.576$  for level correction, with 73% correct sign. It substantially outperforms the linear accounting correction  $\hat{y}^* = y^{EH} - h_i \times 13.64$  ( $\rho = -0.060$ ), capturing nonlinear sector-specific patterns whose partial dependence on hours sits approximately 800 Bs/week above the linear accounting identity.

The welfare implications are direct. Among the 500 diary firms, the poverty headcount ( $P_0$ ) rises from 0.296 under reported income to 0.874 under diary truth — an increase of 58 percentage points. Forty-two percent of firms have negative economic profit.

A simulation exercise confirms that the autoencoder’s failure is not a methodological flaw but an assumption violation. When applied to salaried workers — where the conditional invariance assumption holds by construction — the same autoencoder achieves detection AUC of 0.67–0.68 and rank correlation improvements of 2–5 percentage points. The method is sound; the problem is that self-employment income is generated by a fundamentally different process than wage income.

The paper contributes to three literatures. To measurement error (Bound et al., 2001; Pischke, 1995), it provides the first firm-level validation of ML correction methods against diary-quality ground truth for self-employment income, and formalizes the impossibility of unsupervised correction under structural population differences. To informality (La Porta and Shleifer, 2014; Ulyssea, 2018), it quantifies the welfare cost of the accounting-profit

convention. To ML for economic measurement (Jean et al., 2016; Blumenstock et al., 2015), it shows that method validation requires domain-specific ground truth, not just in-sample fit.

The cost implication is practical: 500 financial diaries at approximately \$80 per firm (\$40,000 total) enable correction of a 14,000-household survey.

The paper proceeds as follows. Section 2 reviews the literatures. Section 3 documents the error anatomy. Section 4 presents the framework, tests the identifying assumption, and derives the impossibility result. Section 5 describes the ML methods. Section 6 describes the data. Section 7 presents the main results. Section 8 examines welfare implications. Section 9 discusses external validity. Section 10 concludes.

## **2 Related Literature**

This paper sits at the intersection of five literatures: measurement error in survey income, machine learning for data quality, informal sector measurement, financial diaries, and poverty measurement under error.

### **2.1 Measurement Error in Survey Income**

Bound et al. (2001) provide the canonical review of survey measurement error, documenting three regularities that have shaped the field: error is mean-reverting (high earners understate, low earners overstate), it varies by income source (self-employment worse than wages, transfers worse than self-employment), and it attenuates regression coefficients toward zero. Pischke (1995), using the PSID validation study, shows that measurement error in earnings attenuates returns-to-education estimates by 10–15%, but the attenuation varies with the income source and survey instrument — suggesting that the classical errors-in-variables framework understates the problem.

For self-employment income specifically, the evidence is worse. Hurst et al. (2014) exploit the gap between reported income and expenditure-based proxies in the Consumer Expenditure Survey to show that the self-employed underreport income by approximately 25%, with underreporting concentrated among high-income households — precisely the pattern that biases poverty measures downward. Meyer and Mittag (2019) link CPS records to administrative tax data and find that measurement error in transfer income generates poverty rate biases of 1–3 percentage points; they note that self-employment income, for which no administrative benchmark exists, is likely subject to even larger errors.

What distinguishes the present paper from this literature is the *source* of the error. Prior work focuses on deliberate underreporting or recall failure — respondents know their income but choose not to report it accurately, or forget. The error documented here is conceptual: respondents report conventional profit accurately (they know their revenue and input costs) but the welfare-relevant measure — economic profit, adjusted for the opportunity cost of the owner’s labour — is a quantity they have no reason to compute and no means to report. This distinction matters for two reasons. First, the error is systematic and predictable from observables (especially hours worked), rather than idiosyncratic — which means it should, in principle, be correctable. Second, it is positive by construction (conventional profit exceeds economic profit whenever the owner works positive hours), which means poverty is always understated, not randomly misclassified.

## 2.2 Machine Learning for Data Quality and Prediction

Autoencoders have been used for anomaly detection in tabular data since Zhou and Paffenroth (2017), who demonstrate that robust deep autoencoders can identify corrupted observations by their high reconstruction error — the logic being that a model trained on clean data will poorly reconstruct observations that deviate from the learned distribution. Isolation forests (Liu et al., 2008) take a complementary approach, identifying observations that are easy to isolate in random partitions of the feature space. Chen and Sathe (2017) show that autoencoder ensembles improve detection stability for high-dimensional tabular data. These methods have been applied extensively to fraud detection, network intrusion, and manufacturing quality control, but applications to economic survey data remain rare — a gap this paper addresses directly.

In development economics, ML has been applied primarily to *prediction* rather than *correction* — a distinction that turns out to be crucial. Jean et al. (2016) use satellite imagery and transfer learning to predict poverty at the village level, achieving  $r^2 = 0.55\text{--}0.75$  against ground-truth survey measures. Blumenstock et al. (2015) predict individual-level wealth from mobile phone metadata in Rwanda, demonstrating that passively collected digital traces contain substantial economic information. McBride and Nichols (2018) use machine learning to improve poverty targeting, achieving substantial reductions in exclusion error relative to traditional proxy means tests.

These papers demonstrate that ML can extract economic information from non-traditional data sources, but they share a common feature: the prediction target and the ground truth are measured in the same units using the same concept. The present paper differs

in a way that turns out to be decisive: I do not aim to predict income from covariates (which would simply reproduce the conditional mean) but to detect and correct the systematic difference between two measures of the *same underlying quantity* that happen to use different accounting conventions. This requires the conditional invariance assumption tested in Section 4 — an assumption that holds trivially for prediction tasks (where training and test data are drawn from the same population) but fails when the training population (salaried workers) and the target population (the self-employed) differ structurally.

## 2.3 Informal Sector Measurement

The difficulty of measuring informal-sector income is a central theme in the development economics literature, closely linked to the debate about the nature of informality itself. Perry et al. (2007) document the heterogeneity of the informal sector, distinguishing between “exit” (voluntary) and “exclusion” (involuntary) motives. La Porta and Shleifer (2014) show that informal firms are typically small, unproductive, and run by poorly educated entrepreneurs, challenging the view of informality as a seedbed of Schumpeterian entrepreneurship. Ulyssea (2018) develops a structural model allowing for both voluntary and involuntary margins, showing that enforcement policies have heterogeneous effects across firm types — a finding that depends critically on accurate measurement of firm profitability.

The measurement challenge for firm profits is especially acute. De Mel et al. (2009) compare four alternative profit measures for Sri Lankan microenterprises — a direct profit question, detailed revenue-minus-cost calculations, inventory changes, and consumption-based proxies — and find that all measures disagree by 30–100%. Their central finding is that the simple direct question performs roughly as well as the detailed alternatives, which they interpret as evidence that measurement error is driven by fundamental difficulty in separating business from household finances, not by question design. McKenzie (2012) pushes this further, showing that measurement error in business profits is so large that standard two-period experimental designs have power of only 0.20–0.40 to detect economically meaningful treatment effects, and arguing for multiple survey rounds to average out the noise.

Within this literature, Grimm et al. (2012) document a finding that is directly relevant to the typology used here: substantial within-sector heterogeneity in West Africa’s informal economy, with “constrained gazelles” — high-potential firms that lack access to capital — coexisting with survival enterprises in the same sectors and cities. The Type C firms in

the companion typology are a related but distinct concept: they appear profitable by the conventional metric but are actually trapped once owner labour costs are accounted for.

The companion papers — Hernani-Limarino (2025a) and Hernani-Limarino (2025b) — develop the theoretical framework and empirical accounting. The first introduces the four-type typology: A (viable:  $\pi^{\text{conv}} > 0$ ,  $\pi^{\text{adj}} > 0$ ), B (failing: both negative), C (trapped:  $\pi^{\text{conv}} > 0$  but  $\pi^{\text{adj}} \leq 0$ ), D (undervalued:  $\pi^{\text{conv}} \leq 0$  but  $\pi^{\text{adj}} > 0$ ). The second provides complete double-entry financial statements validating the diary instrument. The present paper takes the typology and diary data as given and asks whether ML can recover the type classification from survey data alone.

## 2.4 Financial Diaries

The financial diary method — intensive, high-frequency recording of all financial transactions — has generated some of the most detailed evidence on how the poor manage money. Collins et al. (2009) deploy year-long diaries in Bangladesh, India, and South Africa, revealing that even very poor households use a complex portfolio of financial instruments — informal savings groups, moneylenders, shopkeeper credit — and that point-in-time income measures systematically misrepresent their economic lives. Morduch and Schneider (2017) apply the method in the United States, documenting income volatility among moderate-income families that standard surveys miss entirely: households in their sample experienced income swings of more than 25% in five months out of twelve.

In the Latin American context, Maldonado et al. (2017) document a gap of 20–40% between survey-reported and diary-measured business income for Colombian microenterprises, though they do not decompose the gap into its accounting components. The diary used in this paper differs from the Portfolios of the Poor approach in two respects: it is administered by a trained enumerator (rather than self-recorded by the household, which introduces its own biases), and it uses double-entry bookkeeping with explicit tracking of owner hours — providing the accounting framework needed to compute economic profit. The seven-day window is shorter than the year-long Collins et al. diaries but is matched exactly to the EH reference period, enabling firm-level comparison without temporal aggregation.

## 2.5 Poverty and Inequality under Measurement Error

Deaton (1997) provides the foundational treatment of how measurement error biases FGT poverty measures, showing that the direction depends on the error structure: classical



error (mean-zero, independent) tends to raise measured poverty because it spreads the distribution, pushing some observations below the poverty line; but mean-reverting error — exactly the structure documented here — tends to compress the distribution toward the mean, pulling the poor upward and the rich downward, and *reducing* measured poverty. Meyer and Sullivan (2003) argue for consumption-based welfare measures precisely because income measurement error is more severe for the self-employed than for wage workers.

The present paper contributes to this literature by quantifying the bias for a specific, identified source of error — the omission of owner labour costs — rather than relying on estimated or assumed error structures. The diary provides exact measurement of  $\pi^{\text{adj}}$ , enabling computation of the true poverty and inequality measures for the diary sample without the usual caveats about error-structure assumptions.

### 3 The Anatomy of the Error

The measurement error in self-employment income arises from three compounding sources, documented in detail in the companion papers. First, respondents report *conventional profit* — revenue minus purchased inputs — rather than economic profit adjusted for the opportunity cost of their own labour. Second, even conventional profit is measured with recall error: respondents round, forget small transactions, and confuse household and business expenditures, as documented extensively by De Mel et al. (2009) for Sri Lankan microenterprises. Third, the survey reference period (one week in the Bolivian EH) may not be representative of the firm’s annual income cycle, particularly for firms with seasonal demand patterns (McKenzie, 2012).

Define economic profit as  $\pi_i^{\text{adj}} = \pi_i^{\text{conv}} - h_i^{\text{own}} \times w^*$ : conventional profit minus the owner’s hours times the shadow wage. The shadow wage  $w^* = 354$  Bs/week is calibrated to match the typology distribution in the companion paper.<sup>1</sup> The measurement error is  $\varepsilon_i = y_i^{\text{EH}} - \pi_i^{\text{adj}}$ .

Table 1 reports summary statistics by sector. Three properties stand out, each with distinct implications for ML correction.

The error is *large*. The full-sample mean is 1,971 Bs/week, with a median of 1,329. To

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<sup>1</sup>The statutory minimum wage implies  $w^* = 546$  Bs/week ( $= 13.64 \times 40$  hours), but the empirical distribution of shadow wages in the diary sample is lower. Calibration across  $w^* \in [200, 600]$  yields the best typology match at  $w^* = 354$ , consistent with the median urban informal wage. Sensitivity analysis at  $w^* = 400$  and  $w^* = 300$  does not change the qualitative results.

Table 1: Measurement error by sector

| Sector             | $N$        | Mean $\varepsilon$ | Median       | SD           | Skew.        | IQR          | Mean $ \varepsilon $ | $\varepsilon/\pi^{\text{conv}}$ |
|--------------------|------------|--------------------|--------------|--------------|--------------|--------------|----------------------|---------------------------------|
| Comercio           | 182        | 2,918              | 1,581        | 7,806        | 9.94         | 2,320        | 2,991                | 2.79                            |
| Manufactura        | 67         | 1,513              | 1,160        | 1,951        | 1.47         | 1,954        | 1,747                | 0.97                            |
| Transporte         | 114        | 1,320              | 1,063        | 1,631        | 3.16         | 1,161        | 1,411                | 1.09                            |
| Gastronomía        | 54         | 1,465              | 1,200        | 1,245        | 1.58         | 1,071        | 1,470                | 2.18                            |
| Construcción       | 50         | 1,236              | 1,371        | 935          | -0.90        | 1,063        | 1,367                | 2.47                            |
| Servicios          | 33         | 1,871              | 1,868        | 1,351        | 0.42         | 1,813        | 1,919                | 3.06                            |
| <b>Full sample</b> | <b>500</b> | <b>1,971</b>       | <b>1,329</b> | <b>4,910</b> | <b>14.94</b> | <b>1,641</b> | <b>2,067</b>         | <b>1.86</b>                     |

Note:  $\varepsilon = y^{\text{EH}} - \pi^{\text{adj}}$ . All values in Bs/week except  $N$ , Skewness, and  $\varepsilon/\pi^{\text{conv}}$ .

put this in perspective, the mean is roughly five times the poverty line (254 Bs/week per capita). Even the tightest sector — Construcción, where workers are typically paid piece rates that approximate market wages — shows a mean error of 1,236 Bs/week. The error is not a minor statistical nuisance; it is the dominant component of measured income for most firms.

The error is *positive and systematically related to sector*. The error-to-conventional-profit ratio ( $\varepsilon/\pi^{\text{conv}}$ ) averages 1.86 but ranges from 0.97 in Manufactura (where capital costs are relatively high and owner labour is a smaller share of total costs) to 3.06 in Servicios (where the business *is* the owner’s labour). This pattern is economically intuitive: the accounting gap equals owner hours times the shadow wage, and sectors where the owner works more hours — Servicios, Gastronomía, Construcción — have larger gaps relative to conventional profit. Comercio is the outlier: its mean error (2,918) is driven by a few large retail operations with extreme skewness (9.94), while the typical Comercio firm has a more moderate gap.

The error is *heterogeneous within sectors*. Comercio has the widest IQR (2,320 Bs/week), reflecting heterogeneity in firm size and labour intensity. Construcción has the narrowest (1,063), reflecting the relative homogeneity of construction labour. This within-sector variation is important because it determines whether sector-level corrections can be effective: if the error were constant within sectors, a simple sector-mean adjustment would suffice. It is not.

Figure 1 plots  $\varepsilon$  against  $\pi^{\text{adj}}$ , coloured by sector, with a lowess smoother. Two features are immediately visible and have direct implications for the ML methods tested below.

The error is *mean-reverting*: the lowess curve slopes steeply downward, from approximately 6,000 Bs/week at  $\pi^{\text{adj}} = -2,000$  to near zero at  $\pi^{\text{adj}} = 3,000$ . The mechanism is mechanical.

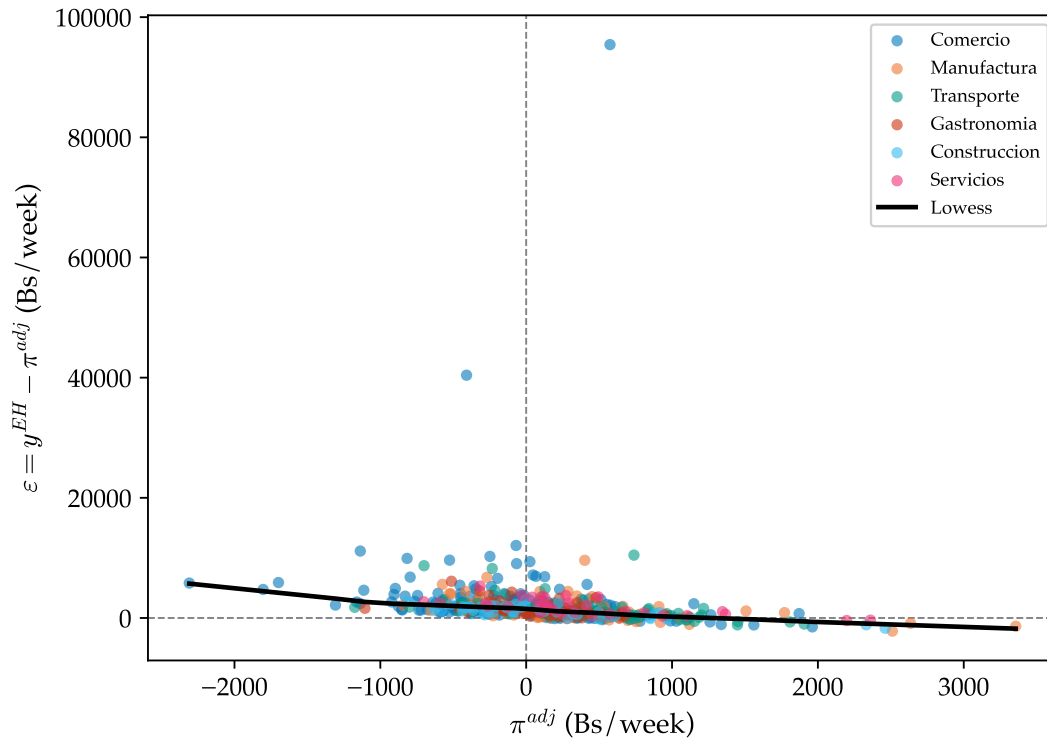


Figure 1: Measurement error  $\varepsilon = y^{EH} - \pi^{adj}$  against diary truth, by sector. The lowess smoother (black) reveals strong mean-reversion: firms with low or negative  $\pi^{adj}$  show the largest errors. Two Comercio outliers exceed 40,000 Bs/week.

Firms with deeply negative economic profit are those whose owner works many hours for little net revenue — so  $y^{\text{EH}}$  (which ignores those hours) is much larger than  $\pi^{\text{adj}}$  (which subtracts them). Firms with high economic profit, by contrast, generate enough revenue that the labour cost is small relative to the total. This mean-reversion is important because it means the error is *negatively correlated with the true value* — exactly the structure that Bound et al. (2001) identify as most problematic for regression analysis.

The error is *heteroskedastic*: Comercio observations dominate the upper tail (two outliers exceed 40,000 Bs/week), while Construcción clusters tightly near the median. Figure 2 makes this clear.

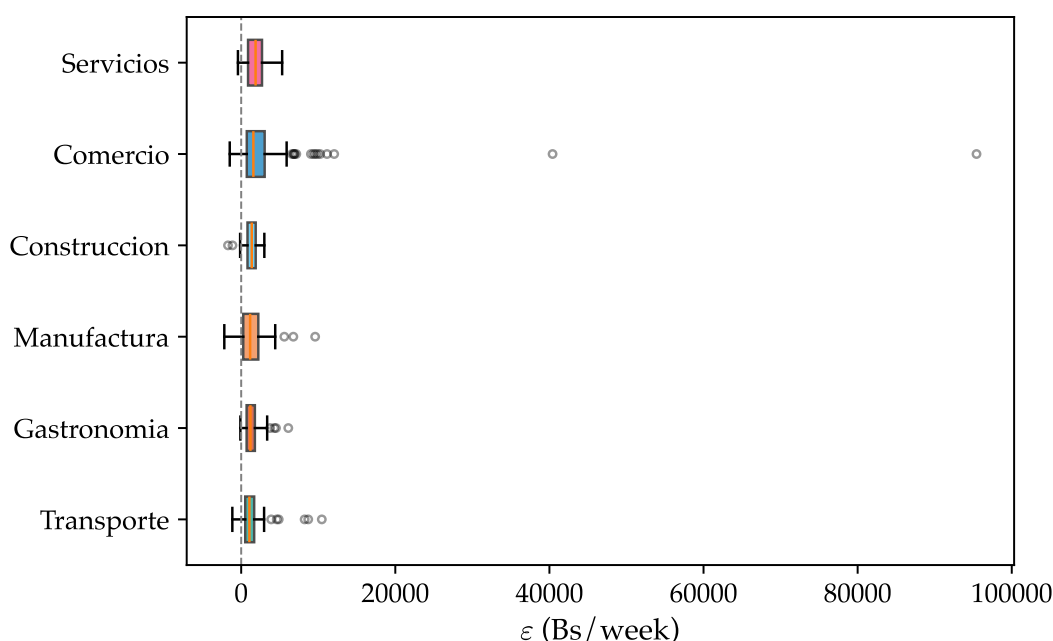


Figure 2: Distribution of measurement error by sector. Comercio has the widest spread and most extreme outliers; Construcción is the most compact.

Taken together, these three properties — positive bias, mean-reversion, and heteroskedasticity — place the error squarely in the non-classical regime. Classical measurement error is mean-zero, independent of the true value, and homoskedastic; under such error, OLS coefficients are attenuated but poverty measures are approximately unbiased (Bound et al., 2001). The error structure here is different in every dimension. The bias is positive (poverty is understated), the correlation with truth is negative (rankings are scrambled), and the variance differs across cells (some groups are more distorted than others). Any ML method that assumes classical error — or that ignores the structural source of the error — will fail.

## 4 Framework

### 4.1 The Measurement Error Model

Let  $y_i^*$  denote the true welfare-relevant income of self-employed individual  $i$ , operationalized as economic profit  $\pi_i^{\text{adj}}$ . The survey reports  $y_i^{\text{EH}} = y_i^* + \varepsilon_i$ , where  $\varepsilon_i$  is the measurement error. Section 3 established that  $\varepsilon_i$  is positive (mean 1,971 Bs/week), mean-reverting ( $\text{Cov}(\varepsilon_i, y_i^*) < 0$ ), and heteroskedastic ( $\text{Var}(\varepsilon_i | X_i)$  varies across sector–gender cells).

An unsupervised correction strategy proceeds as follows. Train a model  $\hat{f}(X)$  on a reference population  $\mathcal{S}$  (salaried workers) where the true income–covariate relationship is directly observable — i.e., where  $y^* = y^{\text{EH}}$  by definition, because salaried workers do not have an accounting gap. Apply this model to the self-employed population  $\mathcal{E}$  to obtain corrected income  $\tilde{y}_i = \hat{f}(X_i)$ . The logic is that if salaried and self-employed workers with the same characteristics would earn the same true income, then deviations of  $y^{\text{EH}}$  from  $\hat{f}(X)$  must reflect measurement error.

This logic requires a key identifying assumption.

**Assumption 1** (Conditional Invariance). *The conditional distribution of true income given covariates is the same for salaried and self-employed workers:*

$$f(y^* | X, \mathcal{S}) = f(y^* | X, \mathcal{E}).$$

Assumption 1 is the analogue of the “common support” assumption in program evaluation: it requires that the income-generating process is the same across populations conditional on observables. This is a strong assumption, and I now show both theoretically and empirically that it fails.

### 4.2 The Compression Result

The following result formalizes what happens when Assumption 1 fails — why the autoencoder achieves large RMSE reduction but near-zero rank correlation. This is the paper’s main theoretical contribution.

**Proposition 1** (Variance Compression under Assumption Failure). *Let  $\hat{f}(X)$  be a consistent estimator of  $E[y^* | X, \mathcal{S}]$  trained on population  $\mathcal{S}$ . Suppose Assumption 1 fails:  $E[y^* | X, \mathcal{S}] \neq E[y^* | X, \mathcal{E}]$ . Define the corrected income for the self-employed as  $\tilde{y}_i = \hat{f}(X_i)$ . Then:*

- (i)  $\tilde{y}_i \xrightarrow{p} E[y^* | X_i, \mathcal{S}] \neq y_i^*$  in general;
- (ii)  $\text{Var}(\tilde{y}) \leq \text{Var}(y_{\mathcal{E}}^*)$  whenever  $\text{Var}(y_{\mathcal{S}}^*) < \text{Var}(y_{\mathcal{E}}^*)$ ;
- (iii)  $\text{Corr}(\tilde{y}, y^*) \rightarrow 0$  when the structural gap  $\Delta_i = E[y^* | X_i, \mathcal{E}] - E[y^* | X_i, \mathcal{S}]$  dominates the covariate-explained variation.

*Proof.* Part (i): consistency of  $\hat{f}$  means it converges to the regression function of its training population, which differs from the target when the assumption fails. Part (ii):  $\text{Var}(\tilde{y}) = \text{Var}(E[y^* | X, \mathcal{S}]) \leq \text{Var}(y_{\mathcal{S}}^*)$  by the law of total variance; when  $\text{Var}(y_{\mathcal{S}}^*) < \text{Var}(y_{\mathcal{E}}^*)$  (as occurs here because  $\mathcal{E}$  includes negative incomes that  $\mathcal{S}$  does not), the corrected distribution is mechanically compressed. Part (iii): decompose  $y_i^* = \tilde{y}_i + \Delta_i + \eta_i$  where  $\eta_i = y_i^* - E[y^* | X_i, \mathcal{E}]$ . Then  $\text{Corr}(\tilde{y}, y^*) = \text{Var}(\tilde{y})^{1/2} / [\text{Var}(\tilde{y}) + \text{Var}(\Delta) + \text{Var}(\eta) + 2\text{Cov}(\tilde{y}, \Delta)]^{1/2} \rightarrow 0$  when  $\text{Var}(\Delta) + \text{Var}(\eta) \gg \text{Var}(\tilde{y})$ .  $\square$

**Corollary 1.** *The corrected income  $\tilde{y}$  can achieve lower RMSE than  $y^{\text{EH}}$  even when  $\text{Corr}(\tilde{y}, y^*) \approx 0$ , provided  $|E[\tilde{y}] - E[y^*]| < |E[y^{\text{EH}}] - E[y^*]|$ .*

The corollary explains the paradox that is central to this paper’s negative result. If the salaried conditional mean happens to be closer in *level* to true economic profit than the reported income is — which it will be whenever the accounting gap pushes  $y^{\text{EH}}$  far above  $y^*$  — then RMSE falls mechanically, even though the corrector contains no information about *which* self-employed workers are rich or poor. The corrector “helps” by pulling everyone toward the centre, not by informing. For welfare analysis — poverty measurement, inequality decomposition, programme targeting — rank preservation matters more than level accuracy, and the corrector fails precisely on this dimension.

### 4.3 Testing Assumption 1

I test Assumption 1 directly by training separate random forest models on salaried workers ( $N = 8,022$ ; 500 trees, max depth 10) and on the diary-observed self-employed ( $N = 500$ ), using the same covariate vector  $X$ . The salaried model predicts  $\hat{y}^{\text{S}}(X) = E[y^{\text{EH}} | X, \mathcal{S}]$ ; the self-employed model predicts  $\hat{y}^{\text{SE}}(X) = E[\pi^{\text{adj}} | X, \mathcal{E}]$ . Under Assumption 1, these predictions should be positively correlated and similar in level when evaluated on the same observations.

Table 2 reports the results. The assumption fails decisively. The correlation between  $\hat{y}^{\text{S}}$  and  $\hat{y}^{\text{SE}}$  evaluated on the 500 diary firms is  $-0.268$  — not merely low, but *negative*. Covariates that predict higher income for salaried workers predict *lower* economic profit

Table 2: Test of Conditional Invariance (Assumption 1)

| Metric                               | Value  |
|--------------------------------------|--------|
| Correlation (RF predictions)         | −0.268 |
| Mean difference (Bs/week)            | 1,955  |
| Median difference (Bs/week)          | 1,817  |
| NN-matched correlation               | −0.162 |
| NN-matched mean difference (Bs/week) | 1,910  |

*Note:* Salaried  $N = 8,022$ , self-employed  $N = 500$ .

for the self-employed. The mean level gap is 1,955 Bs/week, almost exactly equal to the mean measurement error (1,971), confirming that the gap *is* the accounting error. Nearest-neighbour matching (to address potential covariate imbalance between  $\mathcal{S}$  and  $\mathcal{E}$ ) barely changes the results: the matched correlation is −0.162 and the level gap remains 1,910 Bs/week.

Figure 3 visualizes the failure. Both RF models predict approximately flat income–hours profiles, but the salaried model sits approximately 150 Bs/week above the self-employed model across the entire hours distribution. The gap is constant — it does not narrow at high or low hours — which means it cannot be corrected by any model that conditions on the same covariates.

The economic interpretation is straightforward. The accounting gap  $\varepsilon_i \approx h_i^{\text{own}} \times w^*$  is a function of owner hours, which are correlated with the covariates  $X$  that both models condition on. Within any sector–hours cell, salaried workers earn their wage while self-employed workers earn  $\pi^{\text{conv}} - h^{\text{own}} \times w^*$ . The gap is systematic, not idiosyncratic, and cannot be detected by any consistency check that conditions on the same  $X$ . This is why unsupervised correction fails: the error is “invisible” to any model that learns only from the covariate distribution, because it arises from a structural difference in how income is generated, not from an anomaly in how it is distributed.

## 5 Methods

### 5.1 Isolation Forest

The isolation forest (Liu et al., 2008) detects anomalies by recursively partitioning the feature space with random splits. Observations requiring fewer splits to isolate receive higher anomaly scores. I train on the pooled sample of salaried and self-employed workers from the EH, using 17 features: log income, six sector indicators, hours worked, education

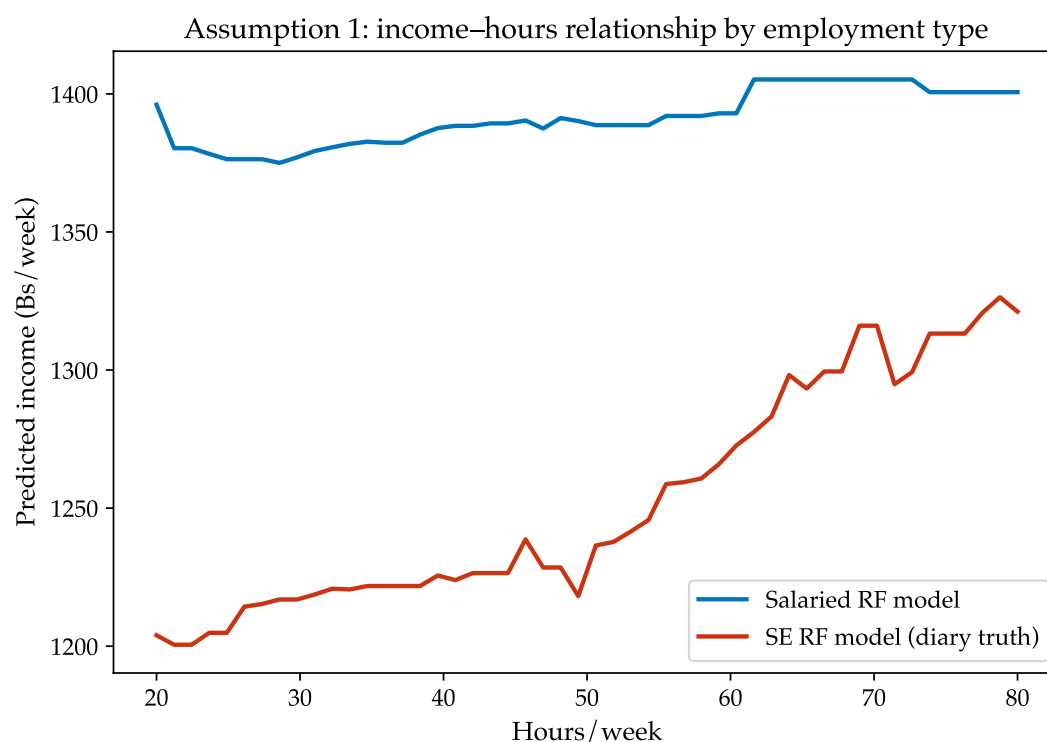


Figure 3: Assumption 1 test: predicted income–hours relationship for salaried (blue) and self-employed (red, diary truth). Both profiles are approximately flat, but the salaried model sits  $\sim 150$  Bs/week above, reflecting the accounting gap. The gap is constant across hours, confirming that the structural difference cannot be absorbed by conditioning on  $X$ .



years, age, household size, number of working-age members, number of children (0–15), asset index (PCA of durable goods), housing quality index, consumption, and gender.

I conduct a grid search over the contamination parameter ( $c \in \{0.05, 0.10, 0.15\}$ ), the number of estimators ( $n \in \{100, 300, 500, 1000\}$ ), and the subsample size ( $m \in \{128, 256, 512\}$ ), yielding 36 configurations. I additionally compute SHAP values (Lundberg and Lee, 2017) for the income feature to isolate its anomaly contribution (“IF Income SHAP”). Full results are in Appendix A.

## 5.2 Autoencoder

The autoencoder is a fully connected neural network with a symmetric encoder–decoder architecture trained to reconstruct its input through an information bottleneck.

*Architecture.* The encoder maps the 17-dimensional input through two hidden layers (128 and 64 units, ReLU activations) to a bottleneck of dimension  $d$ . The decoder mirrors the encoder (64 and 128 units) and maps back to input dimension. Dropout ( $p = 0.2$ ) is applied after each hidden layer. Training uses salaried workers only ( $N = 8,022$ ); the Adam optimizer (learning rate  $10^{-3}$ ); 200 epochs with early stopping (patience 20, monitoring validation loss on a held-out 20% of the salaried sample).

*Loss function.* The loss is a weighted mean squared error:

$$\mathcal{L} = \frac{1}{N} \sum_{i=1}^N \left[ w_{\text{inc}} \cdot (x_i^{\text{inc}} - \hat{x}_i^{\text{inc}})^2 + \sum_{j \neq \text{inc}} (x_{ij} - \hat{x}_{ij})^2 \right], \quad (1)$$

where  $w_{\text{inc}}$  upweights income reconstruction relative to other features. I search over bottleneck dimension  $d \in \{4, 6, 8\}$  and income weight  $w_{\text{inc}} \in \{1, 2, 5, 10\}$  (12 configurations; Table 11). The best configuration is  $d = 4$ ,  $w_{\text{inc}} = 5$  (AUC-ROC = 0.613).

*Detection.* Two anomaly scores: income-specific reconstruction error  $e_i^{\text{inc}} = (x_i^{\text{inc}} - \hat{x}_i^{\text{inc}})^2$  and total reconstruction error  $e_i^{\text{total}} = \sum_j (x_{ij} - \hat{x}_{ij})^2$ .

*Correction.* The reconstructed income  $\hat{y}_i^{\text{AE}} = \hat{x}_i^{\text{inc}}$  serves as the corrected value. Because the AE is trained on salaried workers,  $\hat{y}_i^{\text{AE}}$  estimates  $E[y^{\text{EH}} \mid X_i, S]$  — what a salaried worker with the same covariates would earn. By Proposition 1, when Assumption 1 fails, this compresses the self-employed income distribution toward the salaried conditional mean.

### 5.3 Gradient Boosted Regressor

For the supervised benchmark, I train a gradient boosted regression tree ensemble (scikit-learn `GradientBoostingRegressor`; 200 trees, max depth 5, learning rate 0.1, 5 features per split) on the 500 diary firms, with the measurement error  $\varepsilon_i = y_i^{\text{EH}} - \pi_i^{\text{adj}}$  as the outcome.<sup>2</sup> Five covariates:  $\log y_i^{\text{EH}}$ , sector code, log tool/equipment value, owner hours per week, and household size. Feature importances show that  $\log y_i^{\text{EH}}$  dominates (importance = 0.794), followed by owner hours (0.150), log tool value (0.048), sector (0.004), and household size (0.004).

The correction is  $\tilde{y}_i = y_i^{\text{EH}} - \hat{\varepsilon}_i^{\text{GRF}}$ . Performance is assessed via 100 stratified train/test splits (80/20), stratified by sector to ensure representation in each fold. Mean holdout  $R^2 = 0.362$  (SD = 0.084). I also compute the quantile regression forest (QRF) median prediction, targeting the conditional median of  $\varepsilon$  rather than the conditional mean — more robust to the heavy right tail.

### 5.4 Benchmarks

The *naive* benchmark uses  $y_i^{\text{EH}}$  directly (assumes zero error). The *linear* benchmark computes  $\hat{y}^* = y_i^{\text{EH}} - h_i^{\text{own}} \times 13.64$ , where 13.64 Bs/hour is the hourly minimum wage. The linear correction is exact under two conditions: that  $y^{\text{EH}} = \pi^{\text{conv}}$  (no recall error) and that  $w^* = 13.64 \times h$  (the shadow wage equals the minimum wage times hours). Neither holds precisely in practice.

## 6 Data

### 6.1 The Household Survey (EH)

The Encuesta de Hogares (EH) is Bolivia’s annual nationally representative household survey, administered by the Instituto Nacional de Estadística (INE). The urban self-employment sample includes approximately 14,222 individuals reporting self-employment income. The survey records income from self-employment (“net income from independent activity”), hours worked, sector of activity (six categories: Comercio, Manufactura, Transporte, Gastronomía, Construcción, Servicios), education, household composition, housing characteristics, and asset ownership.

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<sup>2</sup>I refer to this as “GRF” throughout for brevity, though it is gradient boosting (GBR), not the Athey et al. (2019) generalized random forest. The key distinction is that this is a standard predictive model, not a causal forest.

## 6.2 The Financial Diary

The financial diary was administered to 500 urban non-agricultural self-employed firms across six sectors and nine departments. Each firm was observed for seven consecutive days by a trained enumerator who recorded all business transactions using double-entry bookkeeping. The diary captures revenue (by product/service line), all purchased inputs (by category), owner hours (by day), family labour hours, tool and equipment values, and depreciation. The enumerator manual (17 pages, described in Hernani-Limarino 2025b) standardizes all recording procedures.

The key variable is economic profit:

$$\pi_i^{\text{adj}} = \pi_i^{\text{conv}} - h_i^{\text{own}} \times w^*, \quad (2)$$

where  $\pi_i^{\text{conv}}$  is conventional profit,  $h_i^{\text{own}}$  is owner hours per week, and  $w^* = 354$  Bs/week. The accounting identity  $\pi_i^{\text{conv}} - \pi_i^{\text{adj}} = h_i^{\text{own}} \times w^*$  holds exactly in the data (max residual  $< 10^{-6}$ ).

In the diary sample, mean  $\pi^{\text{adj}}$  is 139 Bs/week (median 106) — dramatically lower than the mean reported income ( $y^{\text{EH}}$ ) of 2,110 Bs/week. Forty-two percent of firms have negative  $\pi^{\text{adj}}$ : their owners’ labour costs exceed their conventional profit. The typology distribution is  $A = 285$ ,  $B = 4$ ,  $C = 149$ ,  $D = 62$ .

## 6.3 Linkage

Each diary firm was simultaneously interviewed by the EH, enabling exact firm-level comparison. The merge uses a unique firm identifier, with duplicates resolved to yield exactly  $N = 500$  linked observations.

# 7 Results

## 7.1 Detection Accuracy

Table 3 reports detection accuracy at three thresholds ( $0.5\sigma$ ,  $1.0\sigma$ ,  $1.5\sigma$ ), where an observation is labelled as having “large error” if  $|\varepsilon_i|$  exceeds the threshold. I report AUC-ROC, AUC-PR (which is more informative for imbalanced classes), and precision/recall at the top-50 cutoff.

The results fall into three tiers that correspond exactly to the degree of supervision.

Table 3: Detection accuracy by method

| Method                    | AUC-ROC      |              |              | AUC-PR | Prec<br>@50 | Rec<br>@50 |
|---------------------------|--------------|--------------|--------------|--------|-------------|------------|
|                           | 0.5 $\sigma$ | 1.0 $\sigma$ | 1.5 $\sigma$ |        |             |            |
| Isolation Forest          | 0.457        | 0.484        | 0.480        | 0.061  | 0.052       | 0.448      |
| IF Income SHAP            | 0.520        | 0.623        | 0.631        | 0.123  | 0.072       | 0.621      |
| AE ( $e_{\text{inc}}$ )   | 0.578        | 0.613        | 0.618        | 0.144  | 0.068       | 0.586      |
| AE ( $e_{\text{total}}$ ) | 0.532        | 0.615        | 0.595        | 0.108  | 0.076       | 0.655      |
| GRF $ \hat{\epsilon} $    | 0.983        | 0.998        | 0.991        | 0.981  | 0.116       | 1.000      |
| Linear residual           | 0.500        | 0.500        | 0.500        | 0.058  | 0.000       | 0.000      |

Note:  $N = 500$ . Thresholds: multiples of SD of  $|\epsilon|$ .

The isolation forest performs *at or below chance*: its best AUC-ROC is 0.484 at 1.0 $\sigma$ , worse than the 0.500 a random classifier achieves. This is not a hyperparameter problem — Appendix A shows that all 36 configurations produce AUCs in  $[0.46, 0.50]$ . The IF genuinely cannot detect the error. The reason becomes clear from the SHAP decomposition: the IF Income SHAP score, which isolates the income-specific anomaly contribution, achieves 0.623 — much better than the overall IF score. This means that whatever signal exists comes exclusively from the income dimension; the multivariate covariate structure adds noise, not signal. The IF detects unusual sector–income combinations, not measurement error.

The autoencoder achieves a modest 0.613 at 1.0 $\sigma$  using the income reconstruction error. This is above chance but far from actionable: at any reasonable threshold, the majority of flagged observations would be false positives (AUC-PR = 0.144). The AE does modestly better than random because large reconstruction errors correlate weakly with large true errors — both tend to occur for firms with unusual income levels. But the AE cannot distinguish between a firm whose income is unusual *because it is misreported* and a firm whose income is unusual *because the firm is genuinely different*.

The supervised GRF dominates: AUC-ROC = 0.998 at 1.0 $\sigma$ , with AUC-PR = 0.981 and perfect recall at the top-50 cutoff. The GRF succeeds because it has access to the ground truth: it learns the specific mapping from observables to error, rather than trying to infer it from distributional properties.

Figure 4 displays the ROC curves. The GRF curve hugs the upper-left corner; the AE sits modestly above the diagonal; the IF falls below it; the linear residual traces the diagonal exactly.

Why does the isolation forest fail so completely? Figure 5 provides the answer. The SHAP

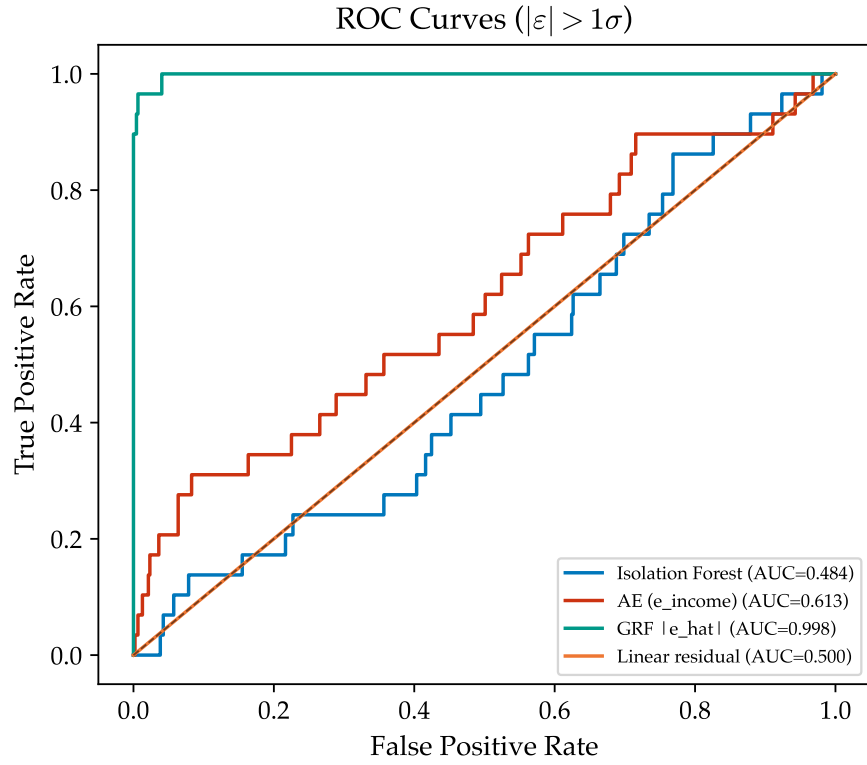


Figure 4: ROC curves at  $|\varepsilon| > 1\sigma$ . The supervised GRF ( $\text{AUC} = 0.998$ ) dominates all unsupervised methods. The IF (0.484) performs below chance.

importance plot shows that sector dummies dominate the IF’s anomaly scoring: the top six features by mean  $|\text{SHAP}|$  are all sector indicators. Income ranks in the middle of the importance distribution, below hours worked and education. The IF is detecting firms that are unusual for their *sector* — a Comercio firm with many children, a Transporte firm with high education — not firms with unusually large measurement error. Since the measurement error is systematic *within* sectors (all Comercio firms overstate income by roughly the same amount relative to economic profit), it is invisible to an anomaly detector that treats within-sector consistency as normal.

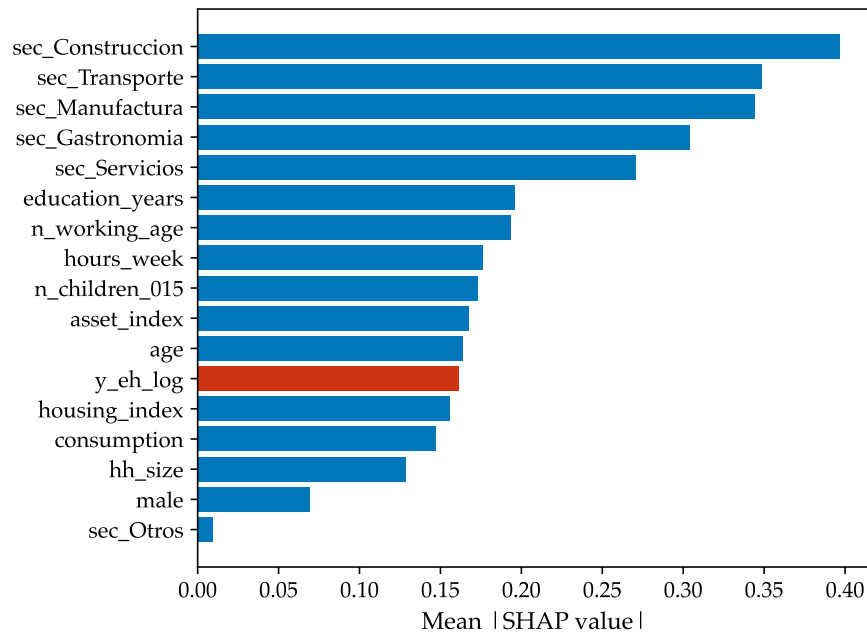


Figure 5: SHAP feature importance for the isolation forest. Sector dummies dominate; income is mid-ranked. The IF detects unusual covariate combinations, not systematic measurement error.

## 7.2 Correction Accuracy

Table 4 reports correction accuracy against diary truth, both overall and by firm type.

The autoencoder produces the paradox predicted by Proposition 1: it dramatically reduces RMSE (from 5,287 to 1,777, a 66% reduction) while achieving near-zero rank correlation ( $\rho = 0.023$ ). The mechanism is exactly as the proposition describes. The AE maps all self-employed incomes toward the salaried conditional mean, which happens to be closer in level to the diary truth than  $y^{\text{EH}}$  is (because  $y^{\text{EH}}$  includes large positive accounting errors). But the AE does not recover the *ranking* of firms — the observation with the highest AE-corrected income is not the observation with the highest true income. It reduces the

Table 4: Correction accuracy against diary truth ( $\pi^{\text{adj}}$ )

| Measure                    | RMSE  | MAE   | Spearman $\rho$ | % Sign |
|----------------------------|-------|-------|-----------------|--------|
| Reported $y^{\text{EH}}$   | 5,287 | 2,067 | −0.178          | 57.8%  |
| AE reconstruction          | 1,777 | 1,388 | +0.023          | 57.8%  |
| GRF corrected              | 3,576 | 695   | +0.576          | 73.0%  |
| QRF median                 | 4,204 | 564   | +0.726          | 81.0%  |
| <i>By firm type:</i>       |       |       |                 |        |
| Type A ( $y^{\text{EH}}$ ) | 5,970 | 1,593 | −0.100          | 100.0% |
| Type A (AE)                | 1,596 | 1,108 | −0.062          | 100.0% |
| Type C ( $y^{\text{EH}}$ ) | 4,529 | 2,737 | +0.015          | 0.0%   |
| Type C (AE)                | 1,942 | 1,708 | +0.223          | 0.0%   |
| Type D ( $y^{\text{EH}}$ ) | 3,493 | 2,727 | −0.131          | 0.0%   |
| Type D (AE)                | 2,161 | 1,963 | +0.000          | 0.0%   |

Note:  $N = 500$ . GRF: mean over 100 stratified 80/20 splits ( $R^2 = 0.362$ ,  $\text{SD} = 0.084$ ). AE correction reduces error for 62% of firms.

average distance to truth by pulling everything toward the centre, not by identifying who is actually rich or poor.

The type-specific results in the lower panel are revealing. For Type A firms (viable,  $\pi^{\text{adj}} > 0$ ), the AE reduces MAE substantially ( $1,593 \rightarrow 1,108$ ) but does not improve rank correlation ( $-0.100 \rightarrow -0.062$ ). For Type C firms (trapped), the AE achieves a positive rank correlation of  $+0.223$  — modest but the only case where the AE provides any rank information. For Type D firms, the AE achieves exactly zero rank correlation, consistent with random assignment.

The GRF achieves  $\rho = 0.576$  with 73% correct sign. The QRF median does even better on rank correlation ( $\rho = 0.726$ ) and sign concordance (81%), though at higher RMSE (4,204). The QRF’s superiority in rank metrics reflects its robustness to the extreme right tail of  $\varepsilon$ , which inflates the GRF’s conditional mean predictions for high-error firms.

Figure 6 displays the correction visually. The left panel plots reported income against diary truth: the relationship is essentially flat ( $\rho = -0.178$ ), with a few extreme outliers exceeding 40,000 Bs/week. The flatness is striking — knowing a firm’s reported income tells you almost nothing about its true economic profit. The right panel shows the AE-corrected income: variance is compressed dramatically (no observations above 12,000 Bs/week), and a slight positive slope emerges for Type C firms (orange), but the overall rank correlation remains near zero. The dashed 45-degree line shows where perfect correction would fall.

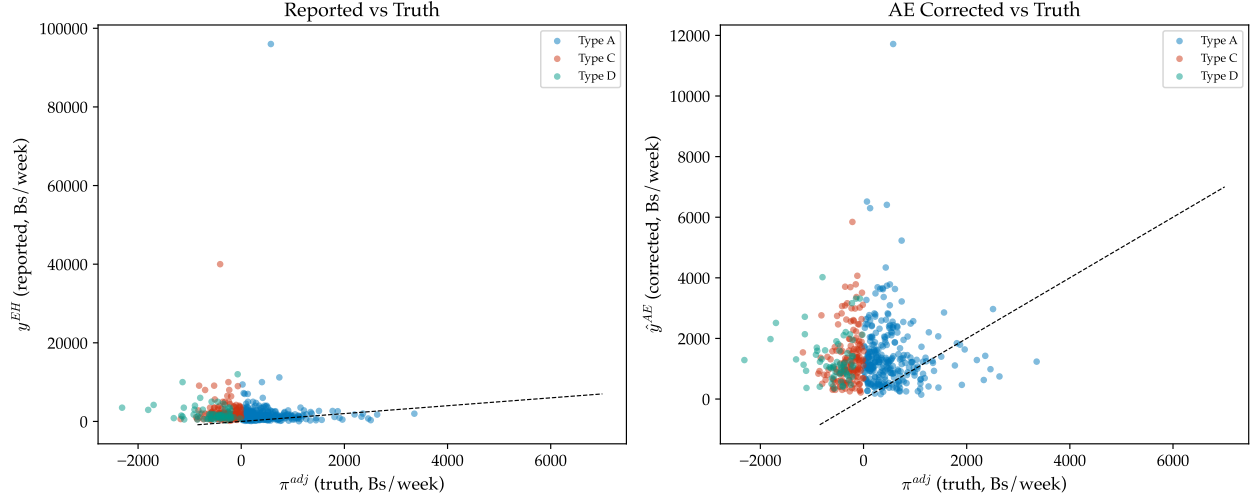


Figure 6: Reported (left) and AE-corrected (right) income against diary truth ( $\pi^{\text{adj}}$ ), coloured by firm type. The AE compresses variance dramatically (Proposition 1) but does not recover rank order. The dashed line marks perfect correction.

### 7.3 Why the AE Compresses but Cannot Correct

Figure 7 provides the detailed diagnostic for the AE’s failure, decomposing its behaviour along four dimensions.

Panel (a) is the most informative. It plots the mean-reversion structure: the AE’s reconstruction error (blue) increases with reported income, while the diary error (red dashed) *decreases*. The slopes run in opposite directions. The economic interpretation is that the AE penalizes high reported income — which looks anomalous relative to salaried workers — while the true error is largest for *low*- $\pi^{\text{adj}}$  firms, where the accounting gap is most severe. The AE and the true error are measuring different things: the AE measures deviation from the salaried distribution, while the true error measures the accounting gap.

Panels (b)–(d) compare error gradient magnitudes across sectors, gender, and hours quintiles. In each dimension, the diary error (red bars) shows substantial variation. Across sectors, Comercio’s mean  $|\varepsilon|$  is 3,000 Bs/week while Construcción’s is only 1,400. Males have larger errors than females (reflecting longer hours in labour-intensive sectors). The top hours quintile has nearly double the error of the second quintile (reflecting the mechanical relationship between hours and the accounting gap). The AE error (blue bars) is flat across all dimensions — it captures at most 5–10% of the true gradient magnitude. The AE cannot see these gradients because they arise from the accounting gap, which is invisible to a model trained on salaried workers where no such gap exists.



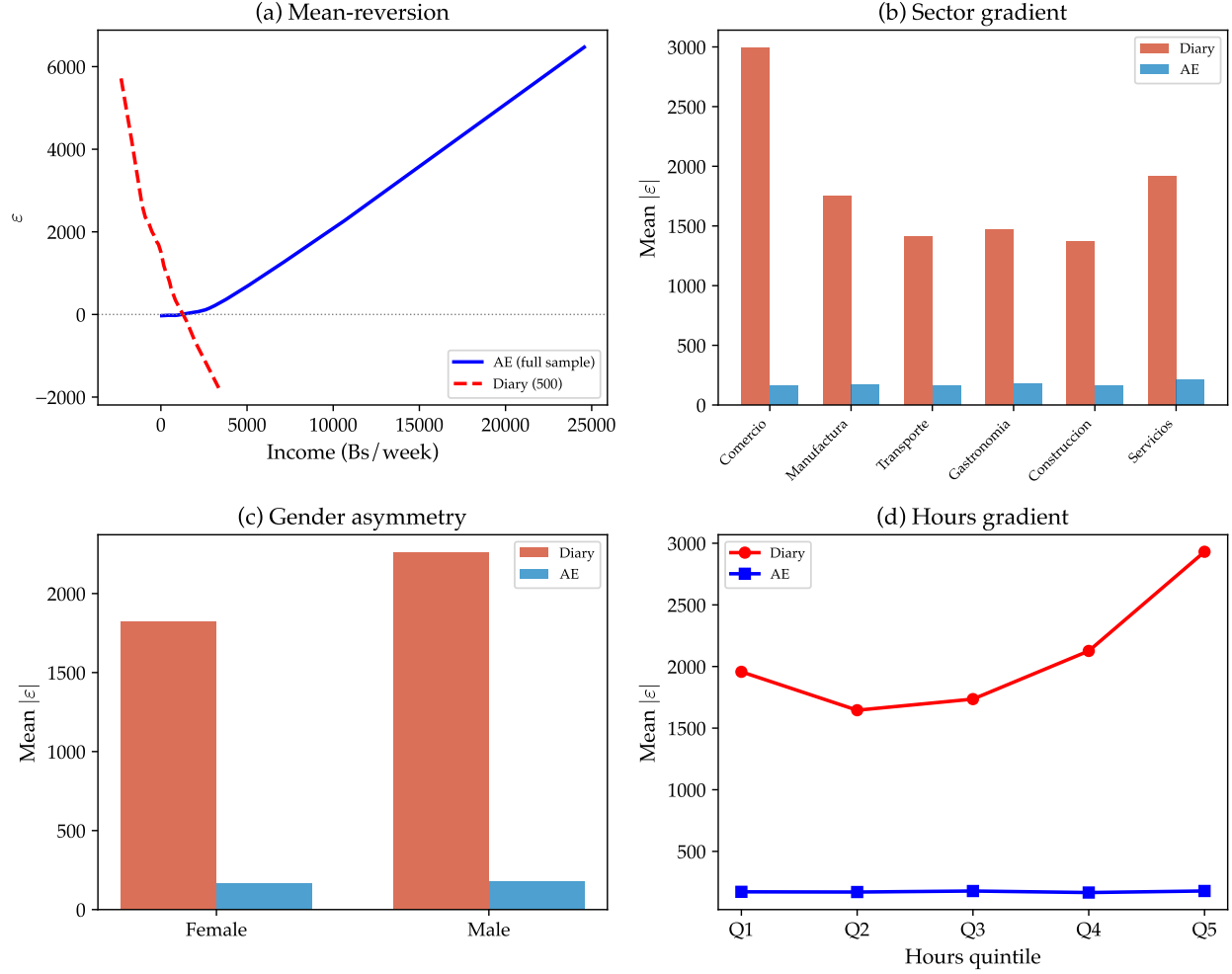


Figure 7: Gradient comparison between AE and diary errors. (a) Mean-reversion slopes run in opposite directions — the AE penalizes high income while the true error penalizes low  $\pi^{\text{adj}}$ . (b) Sector gradient: AE captures <10% of cross-sector variation. (c) Gender asymmetry invisible to AE. (d) Hours gradient invisible to AE.

## 7.4 GRF vs. Linear Correction

The simplest accounting correction available from survey data is  $\hat{y}^* = y_i^{\text{EH}} - h_i^{\text{own}} \times 13.64$ . Table 5 asks: does the supervised GRF add value beyond this simple formula?

Table 5: GRF vs. linear correction

| Method                                        | RMSE  | MAE   | Spearman $\rho$ | % Sign |
|-----------------------------------------------|-------|-------|-----------------|--------|
| Naive ( $y^{\text{EH}}$ )                     | 5,306 | 2,436 | -0.231          | 29.9%  |
| Linear ( $y^{\text{EH}} - h_i \times 13.64$ ) | 5,018 | 1,848 | -0.060          | 39.2%  |
| GRF correction                                | 3,426 | 741   | +0.635          | 79.0%  |

*Note:*  $N = 500$ . Linear:  $\hat{y}^* = y^{\text{EH}} - h_i^{\text{own}} \times 13.64$ .

The answer is decisively yes. The linear correction improves MAE modestly ( $2,436 \rightarrow 1,848$ ) but barely moves rank correlation ( $-0.231 \rightarrow -0.060$ ) — it is still negatively correlated with truth. The linear correction fails because the accounting identity  $\varepsilon_i = h_i^{\text{own}} \times w^*$  is only one component of the total error; the remainder includes recall error in  $\pi^{\text{conv}}$ , temporal mismatch, and sector-specific revenue rounding patterns that the linear formula cannot capture.

The GRF achieves  $\rho = +0.635$  and  $\text{MAE} = 741$ , with 79% correct sign. The improvement is not merely a matter of fitting a more flexible function. Figure 8 decomposes the GRF’s behaviour. The left panel shows true vs. predicted  $\varepsilon$  on holdout data ( $\rho = 0.86$ ): the GRF captures the bulk of the variation in the error, though it compresses the extreme right tail (consistent with regression toward the mean). The right panel plots the partial dependence on owner hours: the GRF curve (blue) sits approximately 800 Bs/week *above* the linear  $h \times 13.64$  line (red dashed) across the full hours distribution. This 800 Bs/week gap represents the non-labour components of the error — recall error, inventory timing, revenue rounding — that the GRF learns from the diary data and the linear formula misses entirely.

Figure 9 confirms that the GRF is well-calibrated: predicted decile medians vs. actual holdout medians lie close to the 45-degree line, with confidence intervals widening in the upper deciles where Comercio outliers dominate.

## 7.5 Reliability Ratios

Table 6 reports the reliability ratio  $\lambda = \text{Var}(\pi^{\text{adj}}) / [\text{Var}(\pi^{\text{adj}}) + \text{Var}(\varepsilon)]$  by sector and gender — the fraction of observed income variance attributable to true income rather than measurement error.

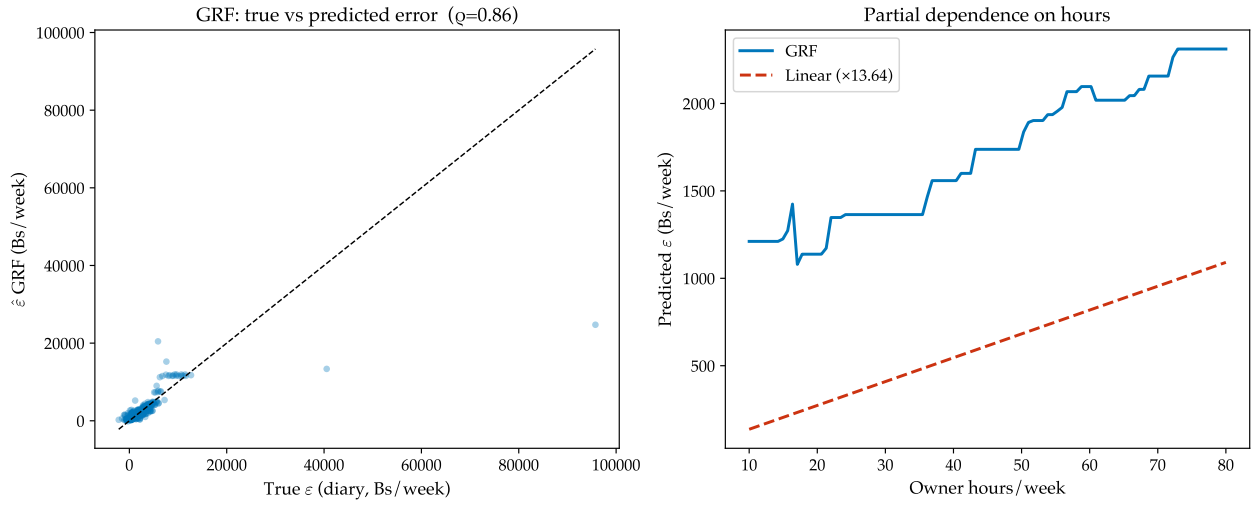


Figure 8: GRF decomposition. Left: true vs. predicted  $\varepsilon$  on holdout data ( $\rho = 0.86$ ). Right: partial dependence on hours — the GRF (blue) sits  $\sim 800$  Bs/week above the linear accounting identity (red dashed), capturing non-labour error components.

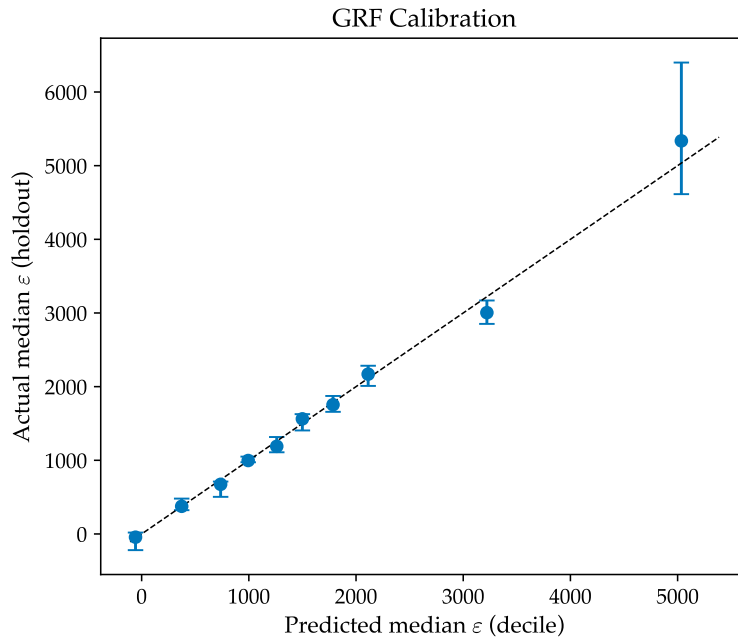


Figure 9: GRF calibration: predicted vs. actual median  $\varepsilon$  by decile on holdout data. Near-perfect calibration, with wider intervals in the upper tail.

Table 6: Conditional error moments by sector and gender

| Sector       | Gender | $N$ | $E[\varepsilon]$ | $\text{Var}(\varepsilon)$ | $\lambda$ |
|--------------|--------|-----|------------------|---------------------------|-----------|
| Comercio     | Female | 130 | 2,053            | 6,044,483                 | 0.066     |
| Comercio     | Male   | 52  | 5,079            | 194,315,977               | 0.001     |
| Manufactura  | Female | 29  | 826              | 2,221,751                 | 0.101     |
| Manufactura  | Male   | 38  | 2,038            | 4,456,631                 | 0.164     |
| Transporte   | Male   | 113 | 1,326            | 2,678,125                 | 0.092     |
| Gastronomía  | Female | 45  | 1,489            | 1,577,813                 | 0.089     |
| Construcción | Male   | 50  | 1,236            | 874,717                   | 0.322     |
| Servicios    | Female | 15  | 1,429            | 1,218,649                 | 0.295     |
| Servicios    | Male   | 18  | 2,240            | 2,114,713                 | 0.170     |

Note:  $\lambda = \text{Var}(\pi^{\text{adj}}) / [\text{Var}(\pi^{\text{adj}}) + \text{Var}(\varepsilon)]$ . Cells with  $N < 10$  omitted.

The ratios are strikingly low and carry a clear economic message. Male Comercio has  $\lambda = 0.001$ : for every unit of variance in observed income, only 0.1% is signal. This is not an abstract statistical quantity — it means that a regression of any outcome on male Comercio self-employment income would capture essentially zero true variation. Even the highest ratio (Construcción males,  $\lambda = 0.322$ ) means that two-thirds of observed variance is noise.

These ratios are an order of magnitude worse than those typically reported for wage income in developed-country validation studies. Bound et al. (2001) report  $\lambda \approx 0.8$ – $0.9$  for hourly wages in the PSID validation study. The self-employment reliability ratios here ( $\lambda = 0.001$ – $0.322$ ) imply that self-employment income in household surveys is closer to pure noise than to a useful economic measure. The practical implication is severe: any regression using self-employment income as the dependent variable — returns to education, poverty correlates, programme evaluation — is dominated by measurement error, with attenuation bias on the order of  $(1 - \lambda) \geq 0.68$  and often exceeding 0.90.

## 7.6 Simulation Validation

To confirm that the autoencoder’s failure on self-employment data reflects an assumption violation rather than a methodological deficiency, I conduct a simulation using salaried workers ( $N = 8,022$ ). I corrupt salaried incomes with four error structures calibrated to match properties of the self-employment error: classical (mean-zero, independent), mean-reverting ( $\varepsilon_i = \alpha + \beta y_i^* + u_i$  with  $\beta < 0$ ), heteroskedastic ( $\text{Var}(\varepsilon_i) \propto y_i^*$ ), and combined. In each case, I train the AE on uncorrupted salaried workers and evaluate on the corrupted sample over 50 replications.

Table 7 shows that the AE works when Assumption 1 holds by construction. Across all four

Table 7: Simulation validation on salaried workers (50 replications)

| Error type      | RMSE    |     | Improv.<br>(%) | Det.<br>AUC | Spearman $\rho$ |       |
|-----------------|---------|-----|----------------|-------------|-----------------|-------|
|                 | Corrupt | AE  |                |             | Corrupt         | AE    |
| Classical       | 390     | 341 | 12.7%          | 0.673       | 0.902           | 0.924 |
| Mean-reverting  | 539     | 474 | 12.2%          | 0.684       | 0.844           | 0.897 |
| Heteroskedastic | 361     | 335 | 7.1%           | 0.663       | 0.928           | 0.927 |
| Combined        | 522     | 475 | 8.9%           | 0.675       | 0.880           | 0.908 |

Note: Salaried  $N = 8,022$ .

error structures, the AE reduces RMSE by 7–13% and achieves detection AUC of 0.66–0.68. Crucially, rank correlations improve meaningfully: from 0.844 to 0.897 for mean-reverting error, from 0.880 to 0.908 for combined error. These are not spectacular improvements, but they are real and in the right direction — confirming that the autoencoder can detect and partially correct income measurement error when the training and target populations share the same income-generating process.

The contrast with the self-employment results is stark and instructive. The simulation AUC (0.67–0.68) is only modestly higher than the self-employment AUC (0.613), suggesting that the AE extracts a similar amount of distributional signal in both cases. But rank correlation collapses from 0.897 (simulation) to 0.023 (self-employment) because the AE is correcting toward the *wrong* conditional mean — exactly as Proposition 1 predicts. The method is sound; the assumption is not.

## 8 Poverty, Inequality, and Typology

### 8.1 Poverty

Table 8 reports FGT poverty measures using a poverty line of 254 Bs/week per capita.

The central finding is dramatic and has direct implications for poverty policy. Among the 500 diary firms, the poverty headcount rises from 0.296 (reported) to 0.874 (diary truth) — an increase of 58 percentage points. This is not a marginal reclassification at the poverty line: the poverty gap ( $P_1$ ) rises from 0.113 to 0.706, indicating that the poor are not merely below the line but deeply below it. The severity index ( $P_2$ ) rises from 0.063 to 0.631, reflecting the concentration of poverty among firms with the most negative  $\pi^{\text{adj}}$ .

These numbers deserve careful interpretation. They do not mean that 87.4% of self-employed workers live in material deprivation — most earn enough *conventional* profit to

Table 8: Poverty measures under alternative income definitions

| Distribution            | $N$    | $P_0$          | $P_1$ | $P_2$ |
|-------------------------|--------|----------------|-------|-------|
| <i>Full EH sample:</i>  |        |                |       |       |
| Reported                | 14,222 | 0.252          | 0.072 | 0.030 |
| AE-corrected            | 14,223 | 0.245 (−0.007) | 0.061 | 0.024 |
| GRF-corrected [cens.]   | 14,222 | 0.979 (+0.727) | 0.768 | 0.654 |
| <i>500 diary firms:</i> |        |                |       |       |
| Reported                | 500    | 0.296          | 0.113 | 0.063 |
| AE-corrected            | 500    | 0.306          | 0.081 | 0.034 |
| Diary truth [cens.]     | 500    | 0.874 (+0.622) | 0.706 | 0.631 |

Note: Poverty line = 254 Bs/week per capita.  $P_1, P_2$ : negative incomes censored at zero. 42% of diary firms have  $\pi^{\text{adj}} < 0$ .

cover household expenses. What they mean is that 87.4% of self-employed workers earn less in economic profit than the poverty line — i.e., they would be better off, in purely pecuniary terms, working for someone else at the prevailing informal wage. This is a statement about the opportunity cost of entrepreneurship, not about material deprivation. But it carries a direct policy implication: social protection programmes that exclude the self-employed on the grounds that they “earn enough” are misidentifying the vast majority of this population.

The AE correction is essentially irrelevant for poverty: it shifts  $P_0$  by less than one percentage point ( $0.252 \rightarrow 0.245$ ) and in the wrong direction — it makes fewer people look poor. This is a direct consequence of Proposition 1: the AE compresses the distribution toward the salaried mean, which is above the poverty line, mechanically reducing measured poverty.

The GRF-extrapolated  $P_0$  for the full sample (0.979) should be interpreted cautiously. The GRF is trained on 500 firms and applied to 14,222; the wide gap between the diary  $P_0$  (0.874) and the extrapolation likely reflects the GRF’s difficulty predicting the right tail of the error distribution for firms outside the diary sample’s support. The diary  $P_0$  of 0.874 is the more reliable benchmark.

## 8.2 Inequality

Table 9 reveals a subtler pattern. For the diary sample, the reported Gini (0.486) slightly overstates the true Gini (0.468) because extreme outliers in  $y^{\text{EH}}$  inflate the upper tail. The AE-corrected Gini (0.352) substantially *understates* the truth — variance compression mechanically reduces measured inequality, as Proposition 1 predicts. This is particularly

Table 9: Inequality measures under alternative income definitions

| Distribution        | Gini  | Theil-L | Theil-T |
|---------------------|-------|---------|---------|
| Reported            | 0.340 | 0.204   | 0.198   |
| AE-corrected        | 0.321 | 0.180   | 0.171   |
| GRF-corrected       | 0.371 | 0.326   | 0.271   |
| Diary 500, Reported | 0.486 | 0.429   | 0.609   |
| Diary 500, AE       | 0.352 | 0.216   | 0.213   |
| Diary 500, Truth    | 0.468 | 0.461   | 0.374   |

problematic because the AE “improves” inequality measurement in the wrong direction: it makes the distribution look more equal than it really is, by pulling everyone toward the mean.

The GRF-corrected Gini for the full sample (0.371) moves in the right direction relative to the reported value (0.340) but remains well below the diary truth (0.468). The Theil indices tell a similar story.

Figure 10 displays the three income distributions, making the scale of the problem visually immediate. The reported distribution (blue) and AE-corrected distribution (red) are nearly indistinguishable — both peak around 1,000–1,500 Bs/week with a long right tail. The diary truth (black dashed) is a fundamentally different distribution: it peaks near 300 Bs/week and has substantial mass below zero. The poverty line (254 Bs/week) and opportunity wage ( $w^* = 354$ ) are marked. The survey is not measuring the same quantity as the diary, shifted by a constant; it is measuring a different quantity entirely.

### 8.3 Typology Recovery

Table 10 reports confusion matrices comparing the survey and AE classifications against diary truth.

Panel A reveals the scale of misclassification. The EH classifies 390 of 500 firms as Type A (viable), when only 285 truly are. Of the 149 Type C firms (trapped), 114 (76.5%) are misclassified as Type A. Of the 62 Type D firms, 49 (79.0%) are misclassified as Type A. The survey’s systematic overstatement of income makes nearly everything look viable.

The AE (Panel B) partially corrects this. It correctly reclassifies 20 additional Type C firms (from 31 to 51), at the cost of 15 additional false positives among Type A firms (from 39 to 54). Panel C summarizes: Type C precision rises from 37.8% to 43.6% (+5.8 pp), recall

Table 10: Typology classification: survey vs. AE-corrected vs. diary truth

|                                    |       | EH    |     |   |    |           |
|------------------------------------|-------|-------|-----|---|----|-----------|
|                                    |       | Truth | A   | B | C  | D   Total |
| <b>Panel A: Diary truth vs. EH</b> | A     |       | 224 | 3 | 39 | 19   285  |
|                                    | B     |       | 3   | 1 | 0  | 0   4     |
|                                    | C     |       | 114 | 0 | 31 | 4   149   |
|                                    | D     |       | 49  | 0 | 12 | 1   62    |
|                                    | Total |       | 390 | 4 | 82 | 24   500  |

|                                              |       | AE    |     |   |     |           |
|----------------------------------------------|-------|-------|-----|---|-----|-----------|
|                                              |       | Truth | A   | B | C   | D   Total |
| <b>Panel B: Diary truth vs. AE-corrected</b> | A     |       | 218 | 0 | 54  | 13   285  |
|                                              | B     |       | 4   | 0 | 0   | 0   4     |
|                                              | C     |       | 88  | 0 | 51  | 10   149  |
|                                              | D     |       | 49  | 0 | 12  | 1   62    |
|                                              | Total |       | 359 | 0 | 117 | 24   500  |

|                                       |   | Type | Method | Precision | Recall | F1    |
|---------------------------------------|---|------|--------|-----------|--------|-------|
| <b>Panel C: Precision, Recall, F1</b> | A |      | EH     | 57.4%     | 78.6%  | 0.664 |
|                                       | A |      | AE     | 60.7%     | 76.5%  | 0.677 |
|                                       | C |      | EH     | 37.8%     | 20.8%  | 0.268 |
|                                       | C |      | AE     | 43.6%     | 34.2%  | 0.383 |
|                                       | D |      | EH     | 4.2%      | 1.6%   | 0.023 |
|                                       | D |      | AE     | 4.2%      | 1.6%   | 0.023 |



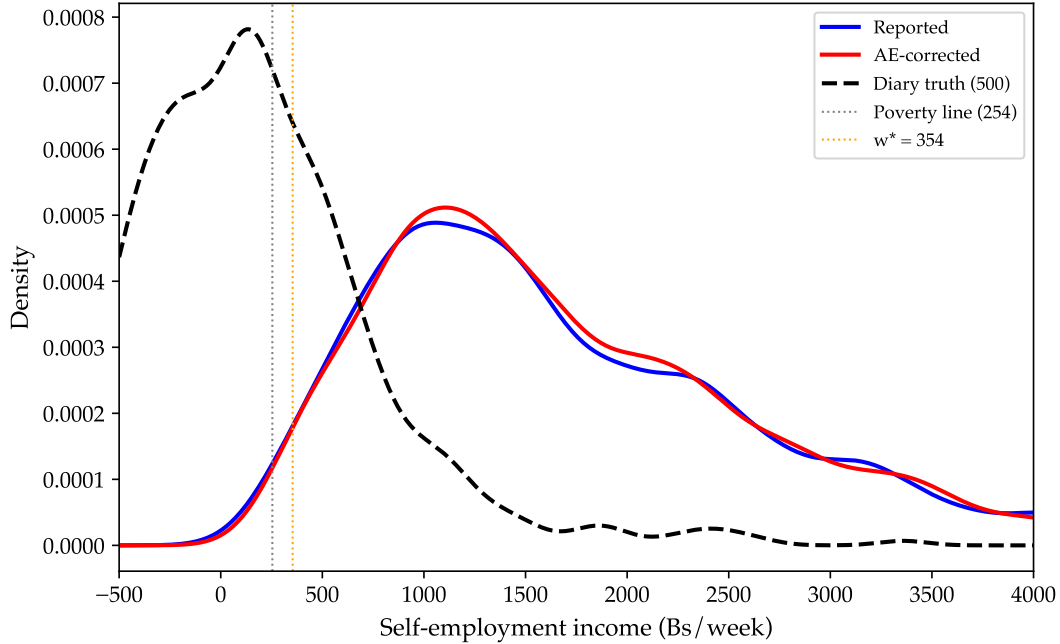


Figure 10: Income distributions: reported (blue), AE-corrected (red), and diary truth (black dashed). The AE-corrected distribution is nearly indistinguishable from the reported. The diary truth is shifted far leftward, with substantial mass below zero. Vertical lines mark the poverty line (254) and  $w^*$  (354).

from 20.8% to 34.2% (+13.4 pp), F1 from 0.268 to 0.383.

The mechanism is variance compression applied to a classification problem. By pulling high reported incomes downward, the AE shifts some firms' conventional profit below  $w^*$ , reclassifying them from A to C. This works imperfectly — it catches firms whose conventional profit was close to the boundary but misses firms deep in Type A territory. Type D remains poorly classified under both methods ( $F1 = 0.023$ ) because Type D firms have negative conventional profit, which neither the AE nor the EH can correct upward.

This is the paper's modest positive result. The AE cannot correct income levels, cannot recover rankings, and makes inequality measurement worse. But it does improve identification of trapped firms — the population most relevant for policy, since these are the workers who would benefit most from wage employment or from capital injections that raise their conventional profit above  $w^*$ . The improvement is not transformative — nearly two-thirds of Type C firms are still misclassified — but it is the best available without diary data.

## 9 External Validity and Cost-Effectiveness

### 9.1 Scope and Portability

The central mechanism — the omission of owner labour costs from survey-reported income — is generic to any setting where self-employed respondents report conventional profit. This includes most household surveys in Latin America and the Caribbean (SEDLAC harmonization), Sub-Saharan Africa (LSMS), and South and Southeast Asia. The *magnitude* of the error will vary with the level of the opportunity wage, the intensity of owner labour, and the sector composition of the self-employed population. In countries where a larger share of the self-employed are in labour-intensive services — street vending, food preparation, transport — the error will be larger; in countries with more capital-intensive self-employment (e.g., agriculture with mechanized equipment), it will be smaller.

The failure of Assumption 1 is likely robust across countries: the income-generating process for self-employment differs structurally from wage employment in all market economies, because the self-employed bear capital risk, labour risk, and demand risk that salaried workers do not. The question is not *whether* the assumption fails but *how much*, and the answer depends on the share of total income attributable to owner labour costs.

The GRF results are less portable. The GRF learns the specific mapping from survey observables to diary-measured error in the Bolivian context. Applying the Bolivian GRF to other countries would require the assumption that the error structure is similar — an untested and likely false assumption. The constructive implication is that each country needs its own diary calibration sample, tailored to its labour market structure and survey design.

### 9.2 Cost-Effectiveness

The 500 financial diaries cost approximately \$80 per firm, or \$40,000 total (enumerator salaries, training, travel, data processing). This enables a correction with  $\rho = 0.576$  and 73% correct sign for the entire 14,222-household self-employment income distribution — a survey that cost orders of magnitude more to administer. The marginal cost of additional observations is roughly constant at \$80/firm, suggesting that expanding to 1,000 firms (\$80,000) would improve performance in thin cells (e.g., Servicios Female,  $N = 15$ ) and enable sector-specific corrections.

The cost-effectiveness calculation depends on the use case. For poverty monitoring —

where the direction and magnitude of the bias matter more than firm-level accuracy — even 200–300 diaries might suffice to calibrate the mean error by sector. For programme targeting — where individual-level classification determines who receives benefits — 1,000 or more may be needed to achieve adequate precision across all sector–gender cells. The optimal diary design is a question for future work, and the reliability ratios in Table 6 provide the variance components needed to compute optimal sample sizes by cell.

## 10 Conclusion

This paper has tested whether machine learning can recover the true income distribution of informal-sector workers from standard household survey data. The answer depends on what “machine learning” means.

Unsupervised methods — isolation forests and autoencoders, trained on salaried workers and transferred to the self-employed — cannot close the gap. The isolation forest performs at or below chance because systematic, within-sector measurement error is invisible to anomaly detection. The autoencoder reduces RMSE by 66% but achieves near-zero rank correlation, compressing all incomes toward the salaried mean without recovering who is actually rich or poor. Proposition 1 shows that this outcome is a necessary consequence of the identifying assumption’s failure: the income–covariate relationship differs structurally between salaried and self-employed workers, and no amount of architectural tuning can bridge this gap without labelled data. One partial exception: the AE improves classification of Type C firms (trapped, with positive conventional profit but negative economic profit), raising precision from 37.8% to 43.6% and recall from 20.8% to 34.2%.

Supervised methods — a gradient boosted regressor trained on 500 diary observations — achieve  $\rho = 0.576$ , 73% correct sign, and  $AUC = 0.998$  for error detection, substantially outperforming the linear accounting identity. The cost of the enabling diary (\$40,000) is small relative to the value of the correction.

The welfare implications are substantial. Among the 500 diary firms, the poverty headcount rises from 0.296 to 0.874 once owner labour costs are accounted for. Forty-two percent of firms have negative economic profit — their owners earn less than they would as wage workers. The majority of firms classified as viable by the survey are actually trapped.

Three implications follow for practice. First, poverty statistics based on self-reported self-employment income are biased downward in countries where the self-employed are a large share of the labour force. The bias is not small: it is 58 percentage points in the diary

sample. Second, unsupervised ML cannot fix this problem — the error is structural, not stochastic, and Proposition 1 shows that correction toward the wrong conditional mean is worse than no correction at all for rank-based welfare measures. Third, a relatively modest investment in financial diaries — 500 firms at \$80 each — can calibrate corrections that substantially improve the accuracy of existing surveys. The diaries do not need to replace the survey; they need to teach the survey what it is missing.

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## A Hyperparameter Sensitivity

### A.1 Autoencoder

Table 11 reports AUC-ROC ( $1\sigma$  threshold) across bottleneck dimensions and income loss weights.

Table 11: Autoencoder AUC-ROC by bottleneck dimension and income weight

| Bottleneck | $w_{\text{inc}}$ |       |              |       |
|------------|------------------|-------|--------------|-------|
|            | 1                | 2     | 5            | 10    |
| 4          | 0.588            | 0.552 | <b>0.613</b> | 0.518 |
| 6          | 0.608            | 0.490 | 0.545        | 0.558 |
| 8          | 0.559            | 0.576 | 0.436        | 0.596 |

Note: Best (bold):  $d = 4$ ,  $w_{\text{inc}} = 5$ , AUC = 0.613.

### A.2 Isolation Forest

Figure 11 displays the heatmap. All AUC-ROC values fall in  $[0.46, 0.50]$ , confirming that the IF's failure is robust to hyperparameter choice.

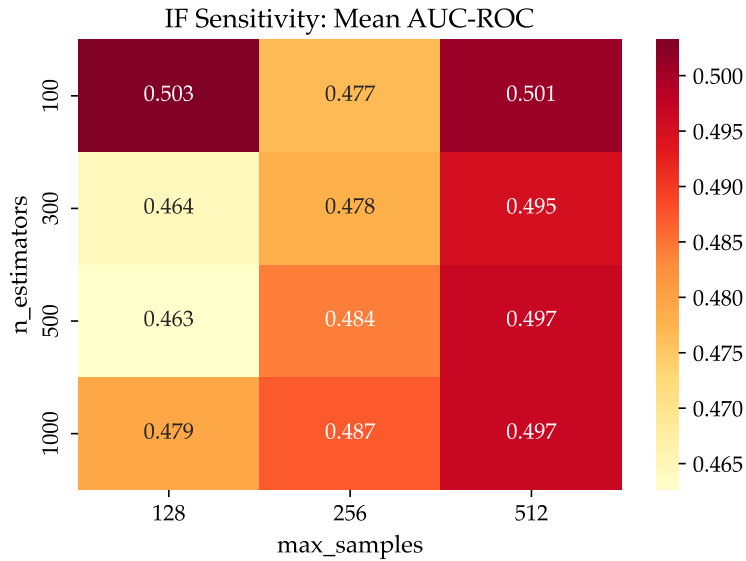


Figure 11: Isolation forest sensitivity: mean AUC-ROC across hyperparameter grid. All values in  $[0.46, 0.50]$ .

Table 12: Isolation forest AUC-ROC by estimators and subsample size

| Estimators | Subsample size |       |       |
|------------|----------------|-------|-------|
|            | 128            | 256   | 512   |
| 100        | 0.503          | 0.477 | 0.501 |
| 300        | 0.464          | 0.478 | 0.495 |
| 500        | 0.463          | 0.484 | 0.497 |
| 1000       | 0.479          | 0.487 | 0.497 |